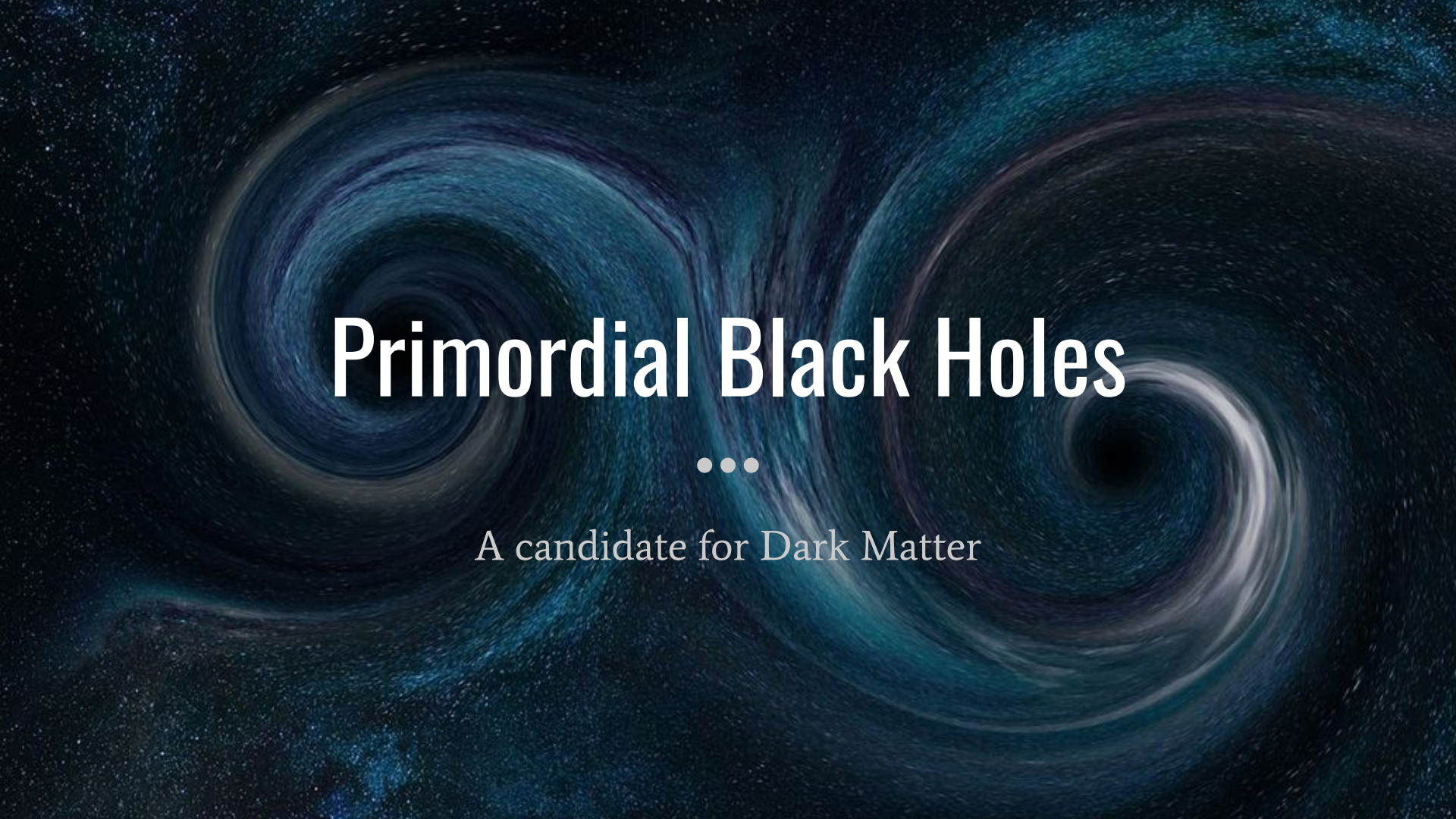


The background of the slide is a dramatic illustration of two black holes in the final stages of a merger. Each black hole is surrounded by a glowing, swirling accretion disk. The disks are primarily dark blue and teal, with some lighter, almost white, regions where the material is being pulled in more intensely. The black holes themselves are represented by dark, circular regions at the center of each disk. The overall scene is set against a dark, star-filled space, with numerous small white dots representing distant stars. The text is centered over the image, with the title 'Primordial Black Holes' in a large, bold, white font. Below the title are three small white dots, and further down is the subtitle 'A candidate for Dark Matter' in a smaller, white font.

Primordial Black Holes

...

A candidate for Dark Matter

DARK MATTER?



DARK MATTER?



1) DARK

DARK MATTER?

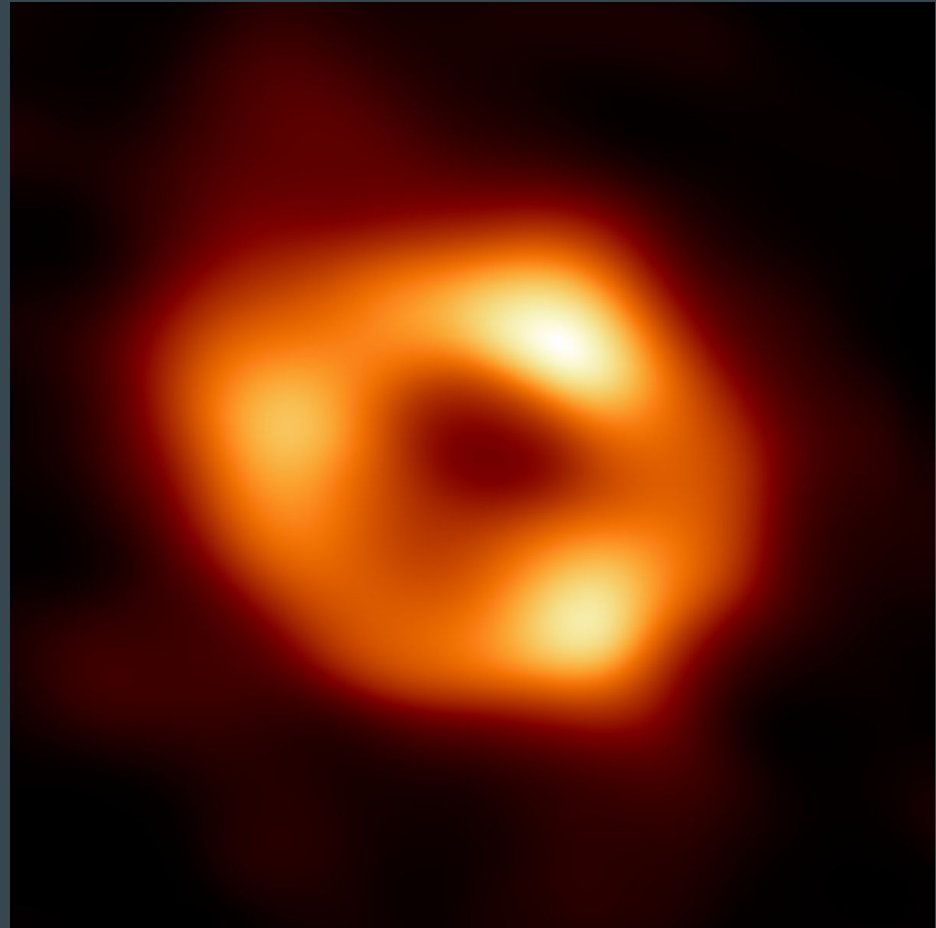


1) DARK

2) MATTER

Black Holes

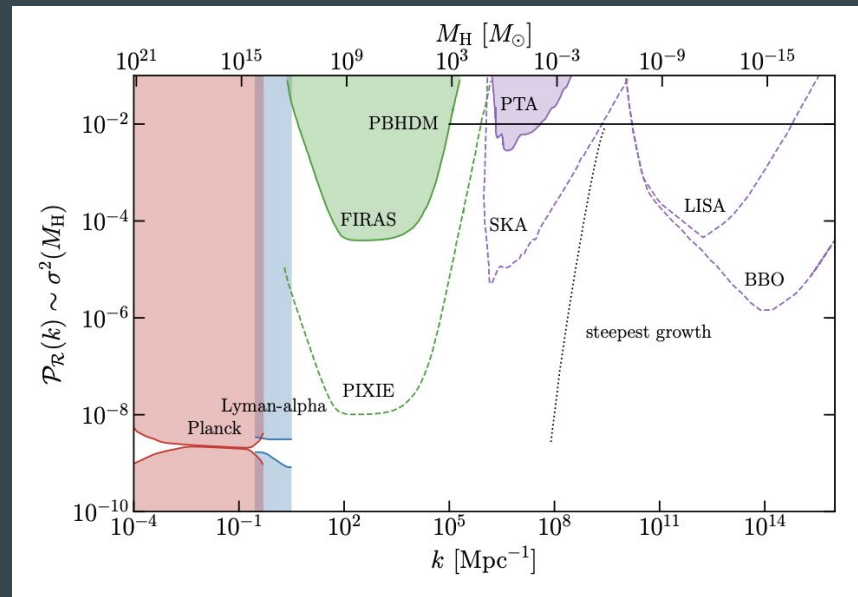
- Supermassive (4 billion $M_{\odot}$)
— TOO BIG :(
- Stellar (over 3 $M_{\odot}$) — NOT
ENOUGH :(
- Primordial — Zel'dovich and
Novikov (1967), Hawking
(1971)



Sgr A*, Event Horizon Telescope, 2022

How are they formed?

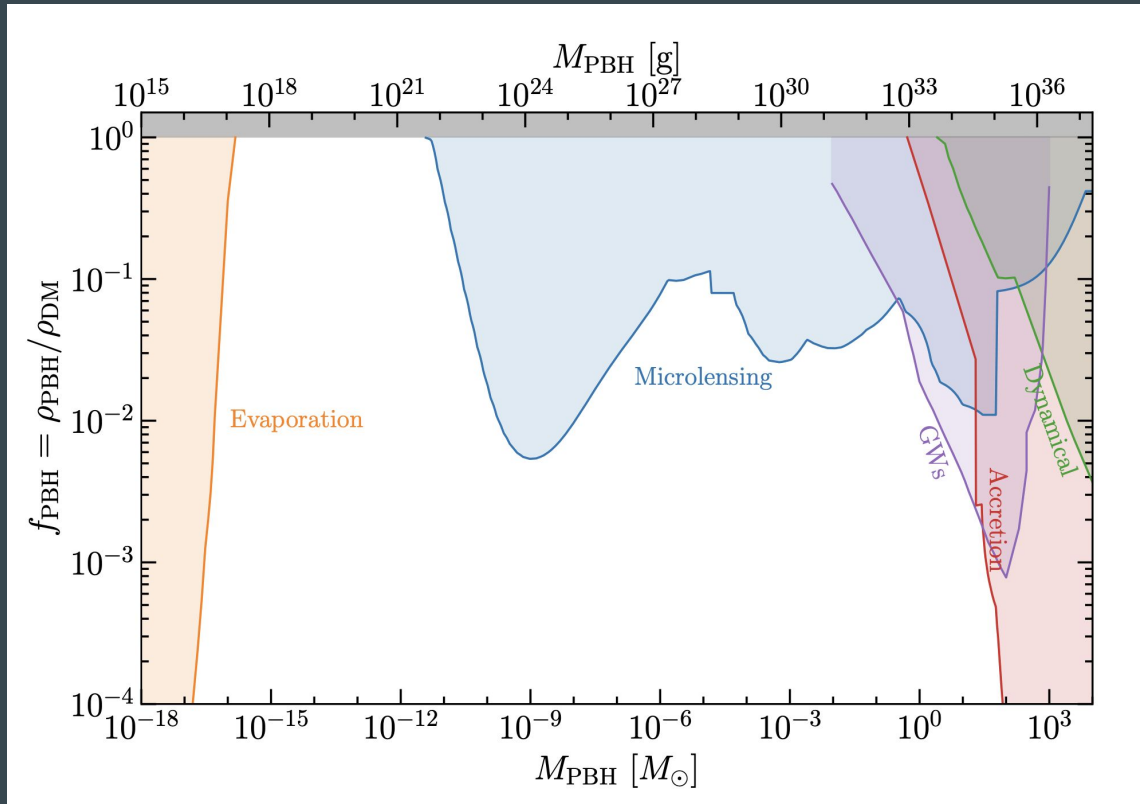
- Collapse of density perturbations in the early universe
- Wide range of possible masses — roughly the horizon mass at the time they were formed
- Also, first order phase transitions



Green, 2024

How are they detected?

- Evaporation
- Gravitational lensing
- Gravitational waves
- Accretion
- Dynamical constraints



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